Introduction

Biomolecular machineries must assemble efficiently to function in a highly heterogeneous physicochemical environment that is shaped by widely varying protein-protein and protein-RNA binding kinetics, molecular abundances, crowding-limited diffusion, and molecule-specific electrical charges ().

These conditions are especially demanding for machineries that process nascent RNAs during transcription because these machineries assemble from many protein factors that bind onto transcribing RNA polymerase II (Pol II).

The cleavage and polyadenylation (CPA) machinery, which cleaves nascent RNAs, adds the poly(A) tail to nascent RNAs, and is tightly coupled with transcription termination is a key RNA-processing machinery that must assemble efficiently.

The CPA machinery assembly must complete within seconds to tens-of-seconds *in vivo* (Chao et al., 1999). This signifies that CPA assembly needs to be very efficient because the transcription process for a human gene of typical length (~25 kilobases) takes minutes to tens-of-minutes given typical transcription elongation rates (2 to 6 kilobases/min) (Darzacq et al., 2007; Fuchs et al., 2014). This assembly efficiency is striking because it is governed by a multitude of protein-protein interactions among CPA factors and between CPA factors and the Pol II carboxyl-terminal domain (CTD). Many protein-RNA interactions between RNA-recognizing CPA factors and 3'-end sequence elements of the nascent transcript, which includes the polyadenylation signal (PAS) and other auxiliary upstream and downstream sequence elements, also prevail ().

Such an efficiency is particularly difficult to explain because the CPA factors involved in PAS recognition (CPSF160, CPSF30, WDR33, and Fip1) are comparable in abundance to or lower in abundance than actively transcribing Pol II (See Result 1, Bekker-Jensen et al., 2017; Jackson et al., 1998; Kimura et al., 2001).

Consequently, it is likely that intense inter-Pol II competition for the limited pool of CPA factors exists within the nuclear environment making efficient CPA assembly and CPA function unlikely.

Experiments also provide insights that efficient CPA assembly will likely be difficult to achieve; experimental perturbations of CPA factor abundance has been shown to alter processing outcomes and shift poly(A) site usage genome-wide (Fusby et al., 2016; Grzechnik & Mischo, 2025; Jin et al., 2025; Kowalski et al., 2024; Mitschka & Mayr, 2022; Tang & Zhou, 2022). Additionally, limiting CPA-factor availability likely reduces the probability of productive CPA assembly on nascent RNAs, thereby increasing the likelihood of delayed or unsuccessful cleavage and polyadenylation (Casañal et al., 2017; Davidson et al., 2014; Kowalski et al., 2024; Neve et al., 2017). Together, these observations indicate that CPA machinery assembly is sensitive to the local availability of factors within the nucleoplasmic pool and that efficient CPA assembly will be even more difficult due to intense inter-Pol II competition for CPA factors.

Such findings sponsor an investigation into whether cells can achieve efficient and reliable CPA machinery assembly within seconds to tens-of-seconds despite the physically constraining conditions of the nucleoplasm, limited factor availability, and competition among elongating Pol II.

In this study, we develop a model that quantifies the efficiency of the CPA assembly dynamics and provides a predictive framework based on assembly efficiency outcomes. Since cleavage,

polyadenylation, and termination each depend on the completion of the CPA machinery assembly, understanding the assembly dynamics is essential to understanding both 3′-end processing and transcription termination.

Methods

A finite rate model of cotranscriptional CPA assembly

We modeled how RNA polymerase II (Pol II) elongation, CTD phosphorylation, CPA-factor loading, PAS recognition, and RNA cleavage jointly determine the distance from a poly(A) signal (PAS) to completed assembly. We represented these processes as a coupled system of finite-rate reactions and transport along a discretized gene.

CTD phosphorylation, CPA-factor loading, and PAS recognition are represented by reversible binding kinetics.

Cleavage removes the Pol II from the modeled template and returns its bound resources to the free pools.

Cleavage is used to quantify the CPA assembly distance, which is the distance travelled downstream of PAS where 50% of CPA machineries have successfully cleaved the RNA transcript's poly(a) site (CAD_{50}).

We can convert this distance into the time it takes for CPA assembly to complete and commit to cleavage at the typical transcription elongation rate.

Gene discretization and polymerase microstates

We represented each gene as a one-dimensional lattice with node spacing $L_a = 100\,bp$. For template length G and PAS at x_{PAS}, we used $N = \lfloor G/L_a \rfloor$ nodes and PAS node $q = \lfloor x_{PAS}/L_a \rfloor$. Recognition and cleavage were permitted from node q through N. The default single-gene template was 25 kb long, with its PAS 20 kb downstream of the transcription start site (TSS).

At node i, $R_i(p, e)$ denotes the population of unrecognizing polymerases carrying p phosphorylated Ser2P sites and e bound E factors, with $0 \le e \le p \le M$. Recognizing polymerases were represented by $REH_i(p, e)$, with $1 \le e \le p \le M$.

In an REH state, exactly one of the e factors is coupled with H.

CTD phosphorylation and E factor exchange

Phosphorylation rate k_{Pon} increased linearly along the gene with rate $k_{Pon,slope}$, $k_{Pon}(i) = k_{Pon,min} + k_{Pon,slope}(i - 1)$.

Dephosphorylation decreased p by one at k_{Poff} only when $p > e$, protecting E-occupied sites.

E associated with either R or REH at $(p - e)k_{Eon}E_f$, reflecting the number of unoccupied phosphorylated sites. Ordinary E - E factor that is not in the recognition complex REH; dissociation occurred at ek_{Eoff} in R and $(e - 1)k_{Eoff}$ in REH. The recognition complex REH dissociates in two ways described below.

PAS recognition and alternative dissociation routes

PAS recognition was modeled as the transition from an elongating Pol II state (RE) to a PAS-bound REH state at the PAS. The effective PAS binding rate was obtained by scaling the basal rate kHon by the mean number of E factors loaded at the PAS. This coupling represents the assumption that Pol II molecules arriving with higher early-factor occupancy are more likely to enter a cleavage-competent, post-PAS complex. At and downstream of the PAS node, each bound E could initiate recognition, giving a transition $R(p,e) \rightarrow REH(p,e)$ at ek_{Hon}.

Recognition complex REH could dissociate via two distinct routes. H dissociation at k_{Hoff} produced $REH(p,e) \rightarrow R(p,e)$, retaining every CTD-bound E factor. Alternatively, the engaged E detached at $k_{Eoff,engaged}$, returning the polymerase to $R(p, e-1)$ and releasing one E. This second route assumed rapid disassembly of the detached EH complex. Because exactly one E was engaged, $k_{Eoff,engaged}$ had no e multiplier. Remaining bound E factors could subsequently support recognition again.

Event	Transition and rate per polymerase
Phosphorylation	$(p,e) \rightarrow (p+1,e)$ at $k_{Pon}(i)$, if $p < M$; in R and REH
Dephosphorylation	$(p,e) \rightarrow (p-1,e)$ at k_{Poff}, if $p > e$; in R and REH
E association	$(p,e) \rightarrow (p,e+1)$ at $(p-e)k_{Eon}E_f$; in R and REH
Ordinary E loss from R	$R(p,e) \rightarrow R(p,e-1) + E_f$ at ek_{Eoff}
Ordinary E loss from REH	$REH(p,e) \rightarrow REH(p,e-1) + E_f$ at $(e-1)k_{Eoff}$
PAS recognition	$R(p,e) \rightarrow REH(p,e)$ at ek_{Hon}, for $i \geq q$
H dissociation	$REH(p,e) \rightarrow R(p,e)$ at k_{Hoff}
Engaged E detachment	$REH(p,e) \rightarrow R(p,e-1) + E_f$ at $k_{Eoff,engaged}$
Cleavage	$REH(p,e) \rightarrow P_f + eE_f$ at k_c

Initiation, elongation, and removal

New polymerases entered node 1 in state $R(0,0)$ at flux $k_{in}P_f$. Elongation advanced a polymerase by one node without changing p or e. R states moved at $k_e = (65\ bp\ s^{-1})/L_a$, corresponding to 65 bp/s, and REH states at $k_{e2} = (30\ bp\ s^{-1})/L_a$, corresponding to 30 bp/s. Reduced velocity was associated with recognition, and unrecognizing polymerases retained k_e downstream of the PAS.

Cleavage removed REH polymerases at k_c. At the final node, R and REH polymerases could also leave by terminal run-off at their respective elongation rates. Both cleavage and run-off returned one Pol II and all e attached E factors to the free pools. Run-off represented escape from the finite simulated window and was not counted as cleavage.

Coupled kinetic equations and resource conservation

Let X collect all R and REH microstate populations, E_f denote free E, and P_f denote free Pol II. The bound-state dynamics were written as

$$\frac{dX}{dt} = (A_0 + E_f A_E)X + P_f b.$$

A_0 contains phosphorylation, dephosphorylation, dissociation, recognition, cleavage, and transport. A_E contains E-association transitions per unit free E. The source $b = k_{in}u$ inflows polymerases into $R_1(0,0)$, where u is a unit vector.

Free E was determined by enforcing conservation of total E. At steady state, the total E pool equals free E plus E bound across all R, RE, and REH states. Thus, the free E concentration used by the fast module was updated until local binding equilibria and global factor conservation were consistent. If w contains the bound-E count e of each microstate and 1 denotes a vector of ones, conservation requires

$$E_{total} = E_f + w^T X, \qquad P_{total} = P_f + 1^T X.$$

Free-pool derivatives were derived from the same reaction operator: $dE_f/dt = -w^T A(E_f)X$ and $dP_f/dt = -1^T A(E_f)X - k_{in}P_f$. Consequently, binding consumes free E, E detachment restores it, initiation consumes free Pol II, and cleavage and run-off recycle both resources.

CPA Assembly Distance

We quantified CPA assembly efficiency using the CPA Assembly Distance (CAD), defined as the downstream distance from the PAS at which the cumulative commitment probability reaches 50%. Conceptually, CAD reports the downstream distance required for half of polymerases that encounter the PAS to commit to CPA. We chose this definition because it maps naturally onto cis-antisense rescue assays, in which protection increases when CPA commitment occurs before downstream interference.

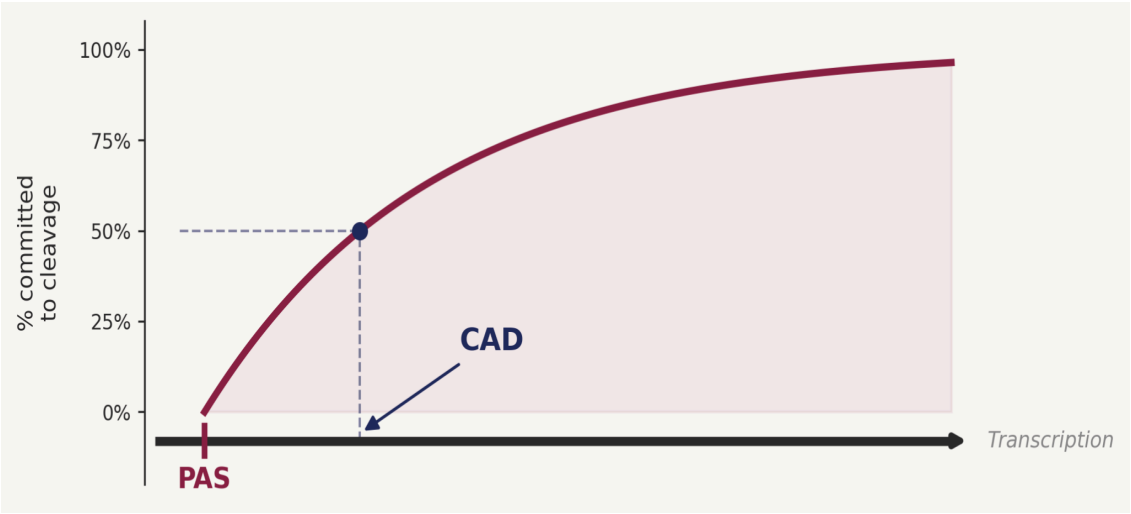


Figure M3. CAD calculation from the commitment flux. The cleavage/commitment flux at downstream node l is k_c · REH(l). CAD is obtained by normalizing the cumulative exit-flux distribution and interpolating the distance at which the cumulative probability reaches 50%.

For each steady-state solution, the local commitment flux was computed as k_c · REH(l). Run-off from the terminal node was treated as a terminal flux to ensure that polymerases reaching the end of the simulated template without commitment were included in the output distribution. The normalized cumulative sum of downstream commitment and terminal fluxes defined the termination CDF F(l), and CAD was obtained by linear interpolation at F(l) = 0.5.

Gene-length and alternative polyadenylation analyses

To test whether gene architecture affects CPA efficiency, we varied the TSS-to-PAS distance while holding the downstream model structure fixed and computed CAD for each simulated gene. To evaluate genome-wide resource competition, we precomputed a three-dimensional lookup grid for R_occupied and E_occupied as functions of free Pol II, free E, and gene length. These functions were interpolated and combined with a log-normal gene-length distribution to solve conservation equations for total Pol II and total E. This procedure links the single-gene assembly model to genome-wide competition for limiting transcription and CPA factors.

To model alternative polyadenylation (APA), we treated the termination CDF downstream of a proximal PAS as the probability that a polymerase commits to that PAS before reaching a distal PAS. Proximal PAS usage was estimated by evaluating the CDF at the inter-PAS distance. PAS strength was varied by sweeping the PAS dissociation parameter, allowing us to ask how intrinsic PAS strength and spacing jointly control proximal-site usage.

Implementation and parameters

Simulations were implemented in MATLAB. The pipeline constructs the E-binding equilibrium rate matrix, solves the steady-state gene-scale ODE system with fsolve, and uses fzero to satisfy E-conservation after each ODE solve. Unless otherwise stated, figures were generated using the default parameter set described in the supplemental methods, which provide full parameter values and script-level reproducibility details.

Results

1. CPA factor abundance motivates/supports a limited-resource model of assembly

To establish the context for CPA-factor recruitment, we compared the abundances of selected CPA factors and RNA polymerase II subunits (Figure 1). The displayed abundance profiles varied substantially across CPA components, with several components less abundant than the Pol II subunits shown. These comparisons motivated a model in which the supply of factors available for CPA assembly is finite.

We represented this supply as a shared pool of effective assembly factors, denoted E, that can bind to elongating Pol II.

- Need to either include a conversion for Pol II core subunit (RPB1) based on the % of actively transcribing Pol II within the cell (based on the total Pol II pool) AND/OR utilize BIONUMBER's actively transcribing Pol II quantity. See Discussion Section.

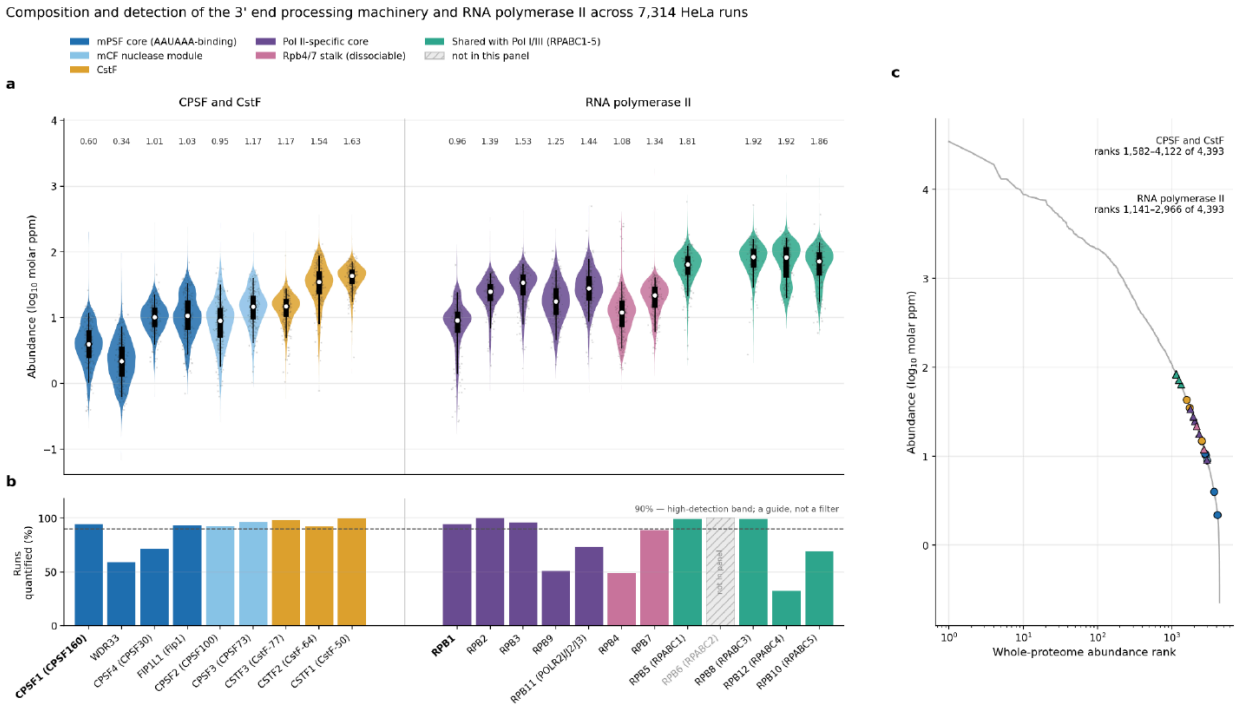


Figure 1. CPA-factor and Pol II abundance comparison.

2. Multiple E-binding sites accelerate CPA commitment

Our kinetic model tests whether multiple binding sites on the Pol II CTD can support efficient CPA assembly under finite factor availability (Figure 2A). The model connects Ser2P-dependent site formation and E loading to reversible PAS recognition and an irreversible commitment step. The CTD capacity parameter M limits both the number of available binding sites and the number of E factors that a polymerase can carry. We quantified assembly efficiency using CAD_{50}, the distance traveled downstream of the PAS by which 50% of the modeled population has committed.

At the reference parameter set, increasing capacity from one to five sites reduced CAD_{50} from 783 to 331 bp, a reduction of approximately 58% (Figure 2B). Further increases in capacity produced progressively smaller reductions in CAD_{50}. The M = 5 commitment curve rose more rapidly over the first 1,000 bp downstream of the PAS when compared to the M=1 commitment curve (Figure 2B-b).

We focused the paired maps on E binding (kEd = kEoff/kEon), PAS recognition (kHd = kHoff/kHon), and irreversible commitment (kc), three controls that strongly affect CAD_{50} (Figure 2C). The maps compare kHd–kEd, kc–kHd, and kc–kEd, with association rates held fixed when varying the binding ratios. M = 5 yielded shorter CAD_{50} than M = 1 wherever both conditions reached 50% commitment. Increasing PAS dissociation prolonged commitment distance, whereas increasing kc shortened it. Greater E dissociation also prolonged CAD_{50} across the E/PAS-binding map.

We used 200–500 bp as a working reference band for efficient commitment within a few hundred base pairs, motivated by the cis-antisense rescue experiments of Chao et al. (1999). We scored $CAD_{50} \leq 500$ bp as meeting the benchmark, including commitment below 200 bp. To assess the joint three-dimensional parameter space, we drew 10,000 independent combinations of kHd, kEd, and kc, sampling each parameter uniformly in log space over the Figure 2C ranges and evaluating the same combinations for both capacities. Association rates and all other parameters were held at their reference values. One site met the benchmark in 23.3% of sampled conditions, compared with 57.0% for five sites. Thus, increasing capacity from one to five sites expanded the fraction of the sampled parameter space supporting commitment within 500 bp by approximately 34 percentage points.

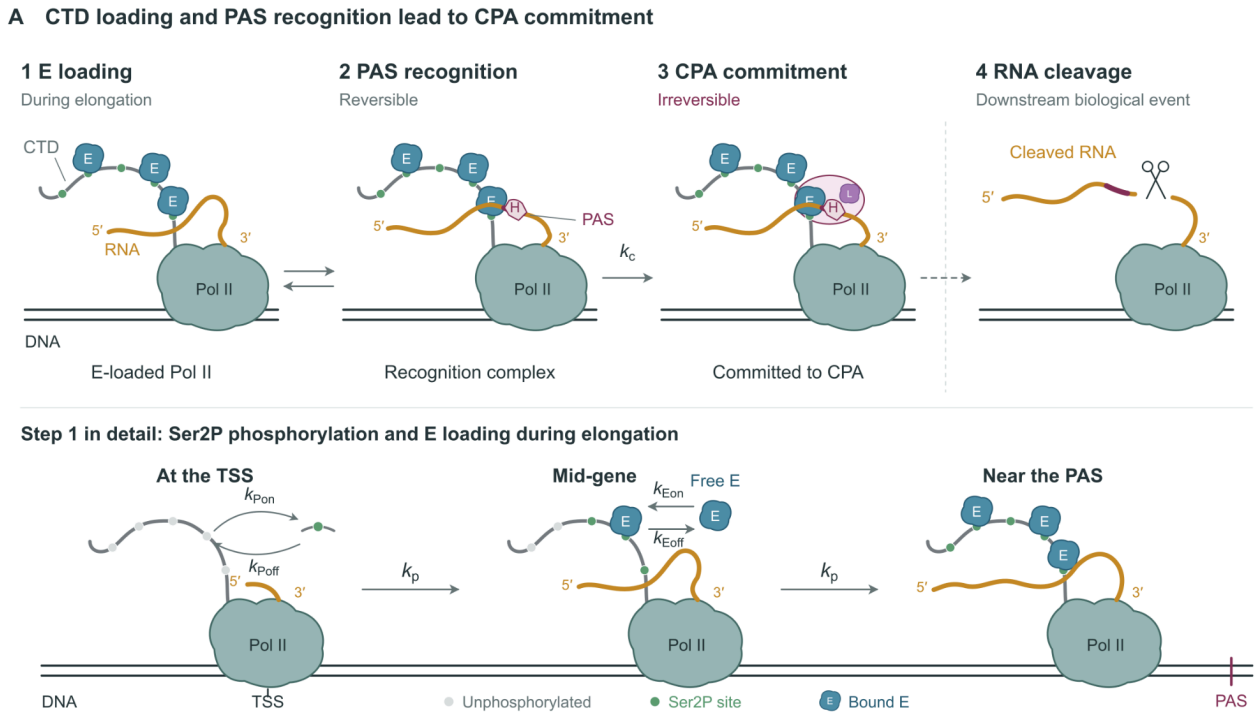


Figure 2A. Model schematic linking E loading, reversible PAS recognition, and irreversible CPA commitment. Cartoon counts and distances are illustrative.

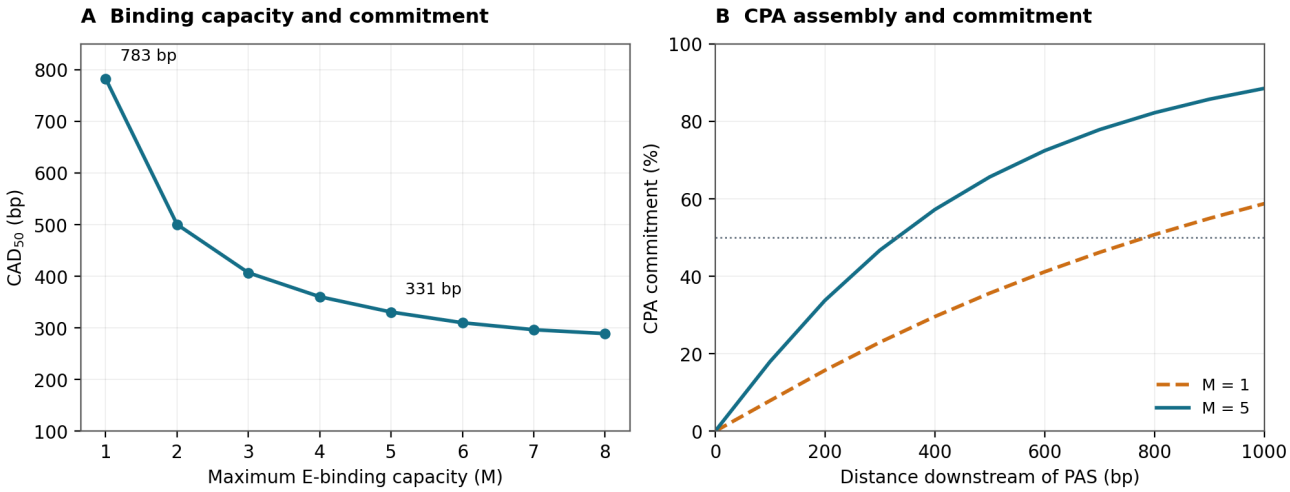


Figure 2B. Binding-capacity dependence of CAD_{50} and commitment curves for M = 1 and M = 5 at the default parameter set. (Note: we can overlay Chao's data later)

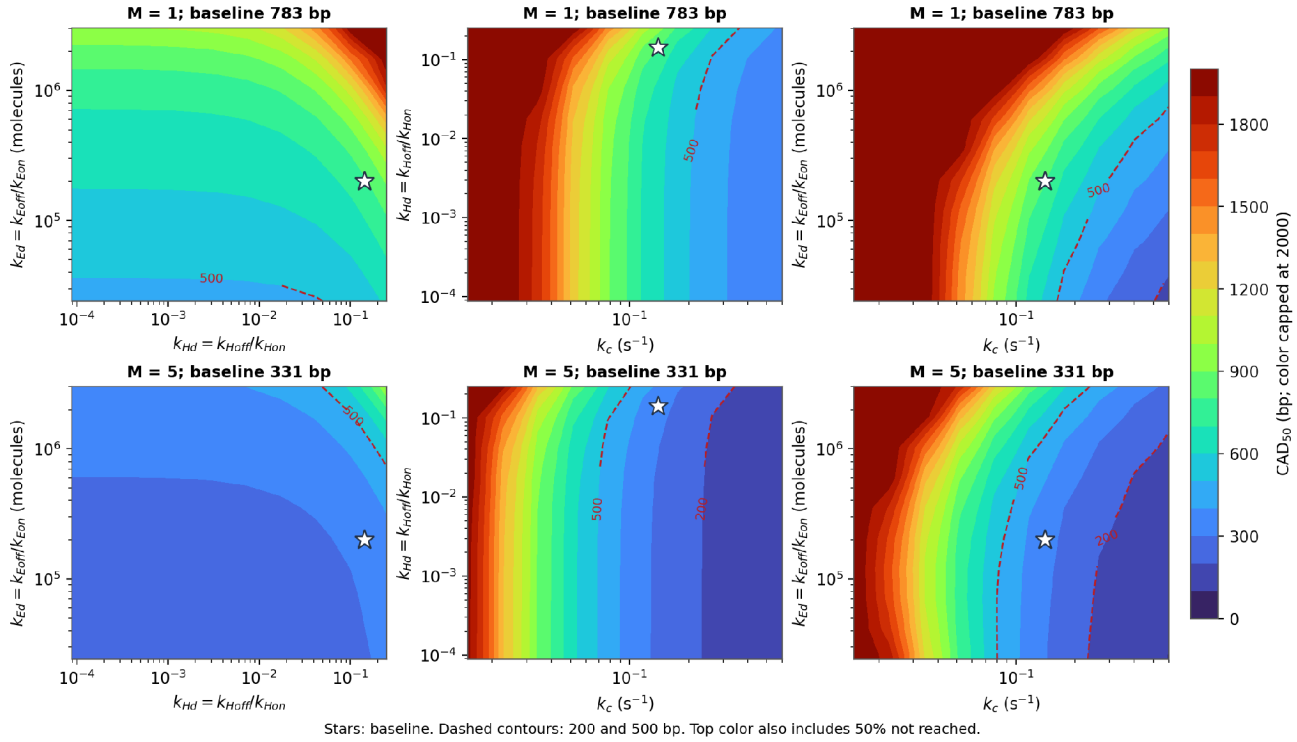


Figure 2C. Paired parameter maps for M = 1 and M = 5. Stars mark the baseline; dashed lines bound the 200–600 bp reference band. Benchmark attainment is scored as $CAD_{50} \leq 600$ bp over all 100 log-spaced conditions in each map, including conditions below 200 bp; an unreached 50% commitment threshold counts as not meeting the benchmark. Displayed colors are capped at 2,000 bp; grey regions and open circles mark an unreached threshold.

3. Multiple binding sites concentrate E loading near the PAS while drawing from a shared factor pool

To examine how multiple sites accelerate commitment, we followed site formation and E occupancy as Pol II approached the PAS (Figure 3). Under the assumed position-dependent phosphorylation kinetics, mean site number and E occupancy increased along the gene. Compared with M = 1, M = 5 polymerases carried fewer E factors near the TSS but more immediately before the PAS, concentrating loading toward the recognition region. Immediately before the PAS, M = 5 polymerases carried an average of 0.89 E factors across 4.44 formed sites, compared with 0.26 E factors across 0.93 sites at M = 1.

Increasing capacity from one to five sites raised the fraction of polymerases carrying at least one E before PAS recognition from approximately 26% to 62%. This fraction measures how many polymerases were preloaded; mean occupancy also counts additional factors on the same polymerase. Increasing M from 5 to 10 raised the preloaded fraction to 73% and reduced CAD_{50} from 331 to 282 bp, showing progressively smaller gains from additional sites.

This local loading draws E factors from a finite shared factor pool (Figure 4A). E retained on one transcribing polymerase reduces the free E available for loading onto others across the gene population. Additional sites therefore increase recruitment capacity while increasing demand on the common pool; their occupancy depends on the E remaining available.

The supplied pool-size sweeps show opposing effects of factor supply and polymerase abundance (Figure 4B). At M=5, increasing total E from 50,000 to 200,000 shortened CAD_{50} from approximately 550 to 260 bp, with smaller reductions at higher supply. Increasing total Pol II from 40,000 to 150,000 increased CAD_{50} approximately linearly, from 335 to 411 bp, at M = 5. These trends are consistent with commitment depending on the balance between factor supply and polymerase demand.

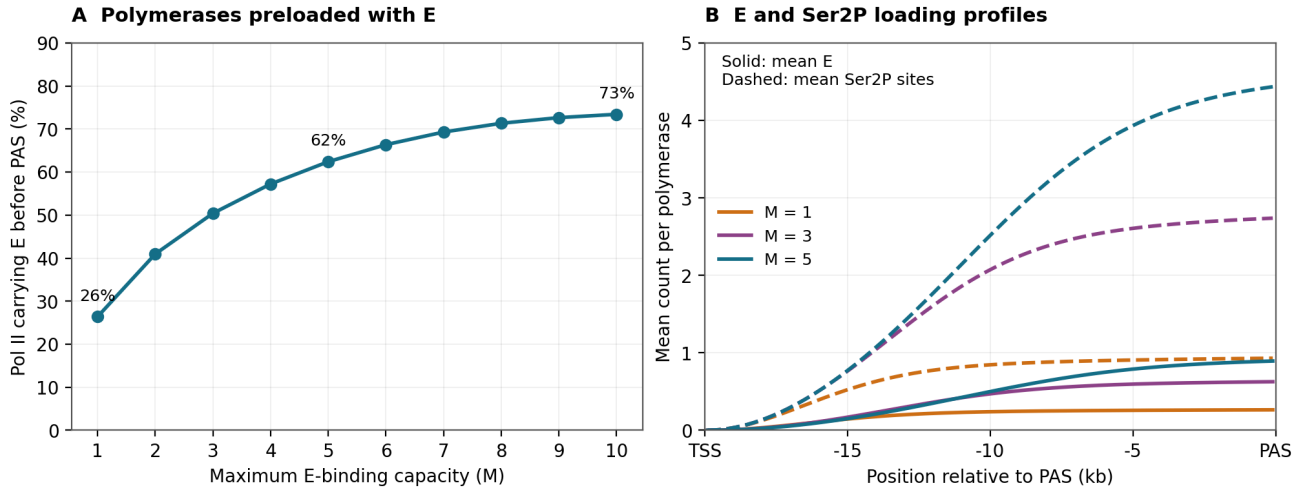


Figure 3. Fraction of polymerases carrying E immediately before the PAS and mean E and Ser2P-site profiles for selected binding capacities. Profiles exclude the PAS-recognized region.

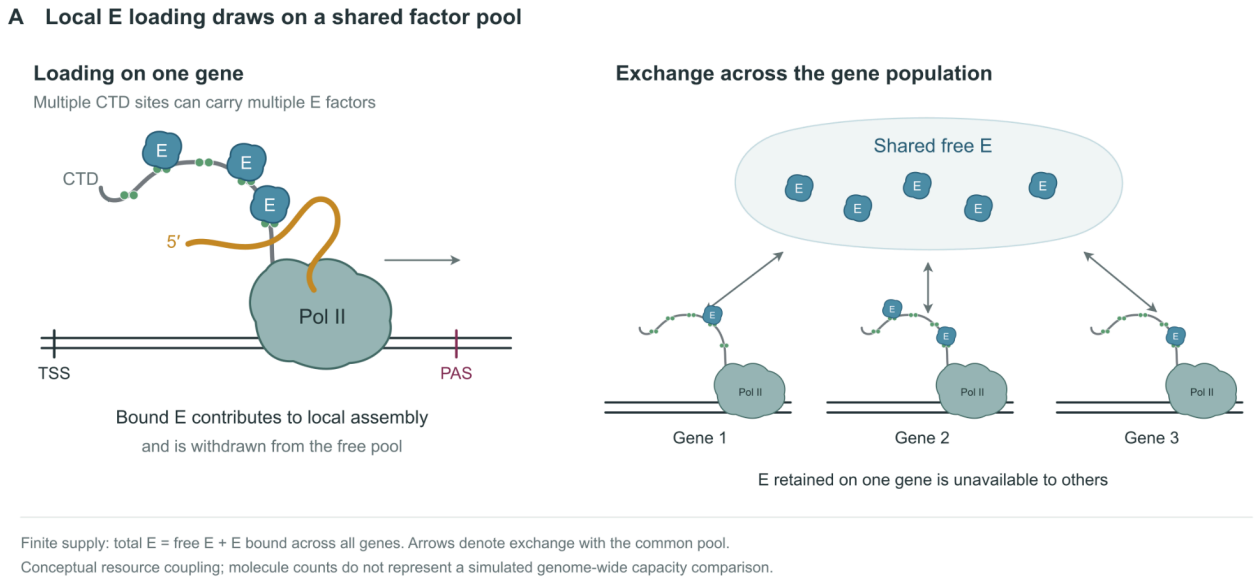


Figure 4A. Conceptual link between local E loading and shared factor availability.

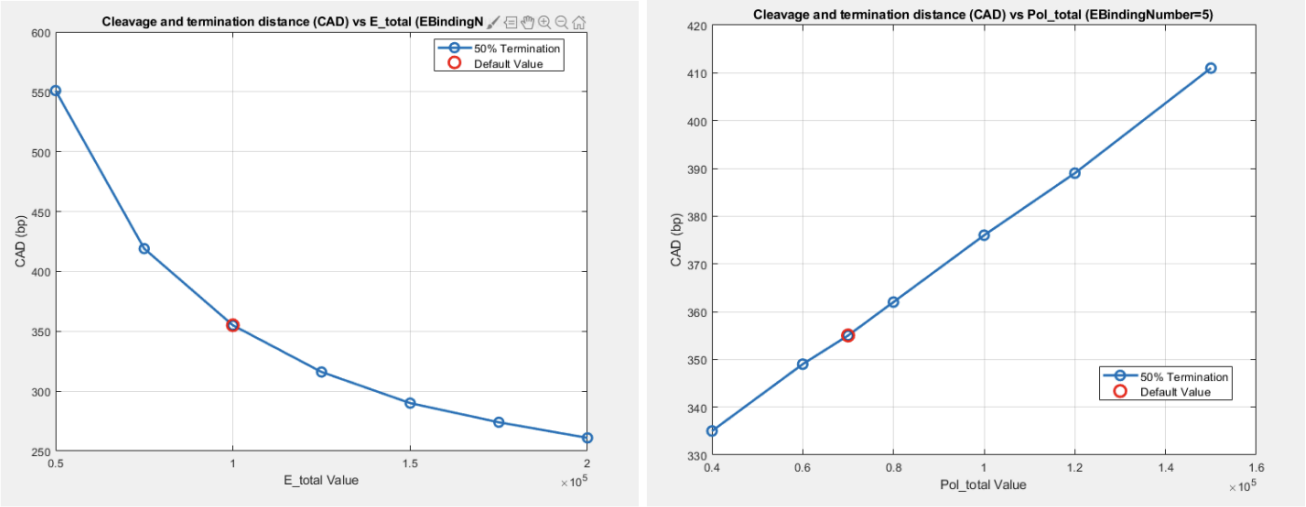


Figure 4B. Supplied pool-size sweeps: increasing total E decreases CAD_{50} (left), whereas increasing total Pol II increases CAD_{50} at M = 5 (right). Red circles indicate the default condition.

4. Gene length limits the opportunity for E preloading

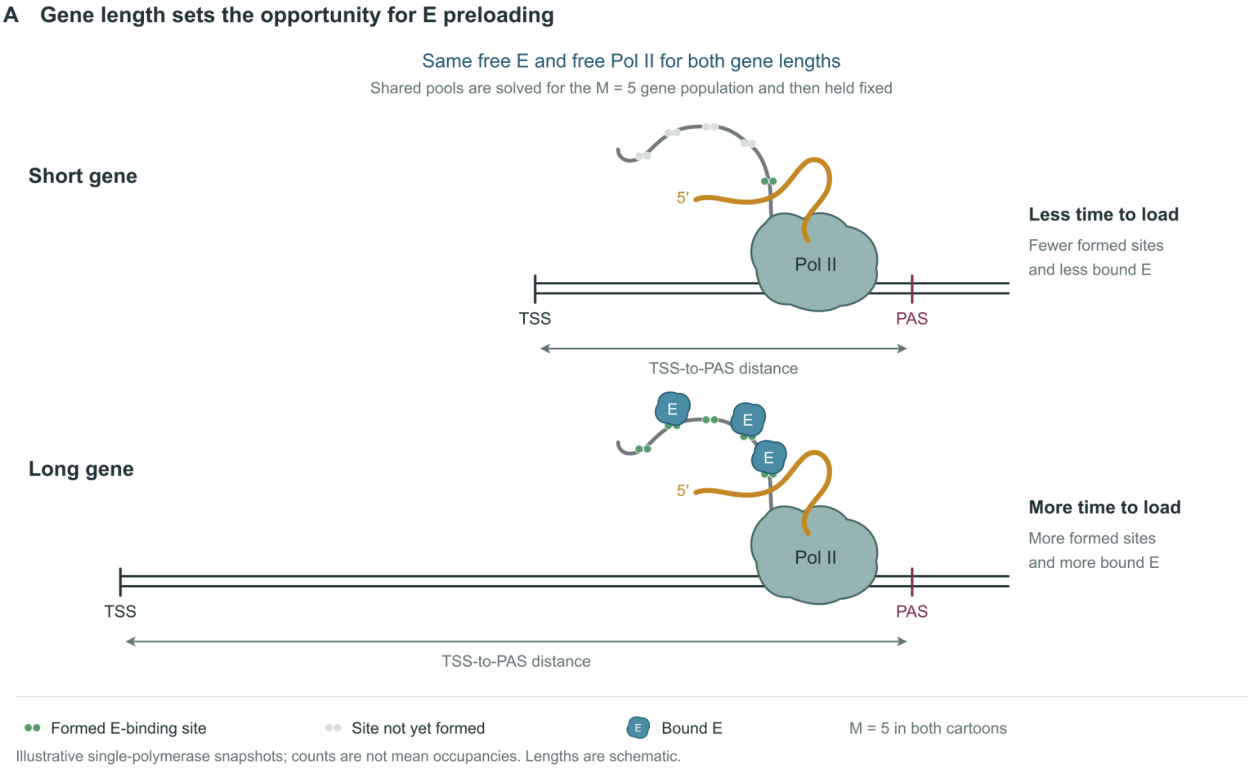
Because E loading develops during elongation, we next examined whether the distance from the transcription start site to the PAS affects commitment (Figure 5A). We first calculated the free E and Pol II remaining after accounting for binding across a population of 10,000 genes with M = 5. We then used those same free pools for every gene length and both binding capacities, so each comparison had the same external factor supply. Short genes reached the PAS with fewer formed sites and less bound E.

For a strong PAS with kHoff = 0.01 s^{-1} and M = 5, CAD_{50} decreased from approximately 3,460 bp at a TSS-to-PAS distance of 2.5 kb to approximately 288 bp at 200 kb (Figure 5B). Genes that are longer than ~10 kb fall within the 200-600 CAD commitment distance. For genes longer than ~20 kb, further reduction of the CAD commitment distance is minimal. In the same free pools, M = 5 produced shorter commitment distances than M = 1, with greater separation between the curves for longer genes. The accompanying profiles showed mean E occupancy immediately before the PAS rising from 0.12 on the 5 kb gene to 0.73 on the 20 kb gene.

To place this length dependence in a human genomic context, we calculated the genomic distance from the annotated transcript start to the transcript 3′ end for one MANE Select transcript per protein-coding gene on chromosomes 1–22, X, and Y (Figure 5C). The 19,229 retained genes had a median span of 22.5 kb, and approximately 47% were shorter than 20 kb.

The observed distribution therefore includes many genes in the length range where the model predicts that commitment depends strongly on preloading opportunity.

MANE reconciles RefSeq and Ensembl/GENCODE transcript annotations, using CAGE and polyadenylation evidence to help define transcript ends. We use the resulting genomic spans, including introns, as a proxy for TSS-to-PAS distance. They end at the annotated transcript 3′ end rather than the upstream PAS hexamer and do not resolve alternative PAS feature usage. NCBI MANE documentation; MANE v1.5 source files.



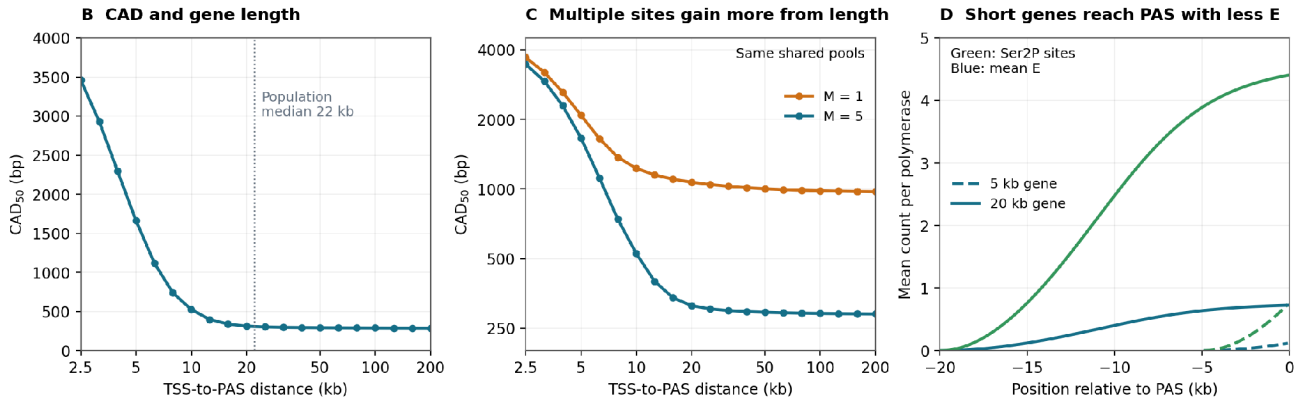


Figure 5A–B. Short- and long-gene schematic, CAD_{50} versus TSS-to-PAS distance, and loading profiles. Green cartoon dots represent formed sites; grey positions have not yet formed available sites. Both capacities use the shared pools solved for M = 5. The 22 kb line marks the assumed population median.

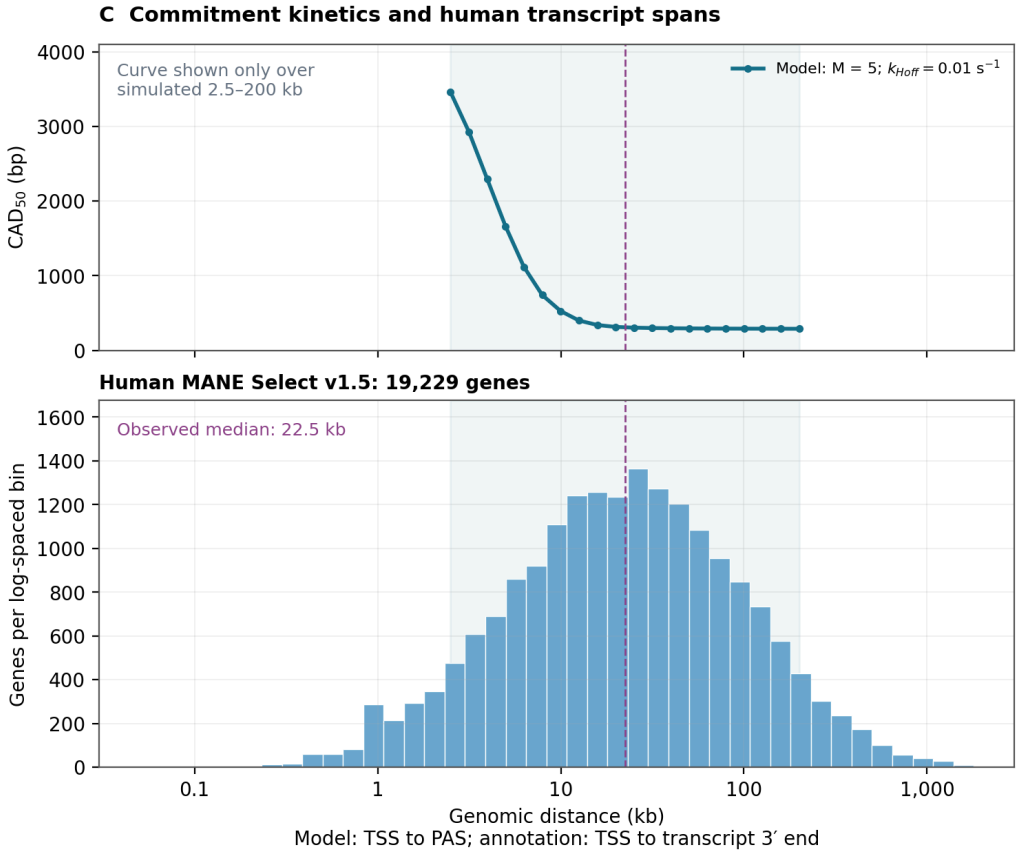


Figure 5C. Human transcript spans compared with the modeled length dependence of commitment. Upper panel: the existing M = 5, kHoff = 0.01 s^{-1} curve, shown only over its simulated range of 2.5–200 kb. Lower panel: the full distribution of genomic TSS-to-transcript-3′-end distances for 19,229 MANE Select protein-coding genes on assembled chromosomes, with one transcript per gene. Both panels share a logarithmic distance axis; shading marks the simulated range, and the dashed line marks the observed median of 22.5 kb. Approximately 85.8% of spans fall within the simulated range; all shorter and longer spans remain in the histogram. Distances are coordinate differences, including introns, verified against the MANE mRNA annotation.

5. Commitment kinetics provide a mechanism for distance-dependent APA regulation

We next considered how the same commitment kinetics could influence the choice between proximal and distal polyadenylation sites. The interval between sites defines the distance over which proximal commitment can occur before the distal PAS is transcribed (Figure 6A). Assuming that reaching the distal site closes this opportunity, the cumulative commitment probability at the inter-PAS distance estimates proximal usage.

To represent differences in PAS recognition, we varied the effective dissociation rate kHoff from 0.001 to 10 s^{-1} while holding kHon fixed; larger kHoff therefore corresponds to weaker effective binding. This broad exploratory range was motivated by reported nanomolar-to-micromolar affinities of PAS-containing RNAs for CPSF/mPSF, which depend on both hexamer sequence and flanking RNA. These affinity measurements motivate the sweep but do not directly determine its individual dissociation rates. Hamilton et al., 2019; Huang et al., 2025.

Proximal commitment increased with inter-PAS distance, while larger kHoff values shifted commitment toward greater distances (Figure 6B). The curves for kHoff = 0.001 and 0.1 s^{-1} nearly overlapped; larger dissociation rates produced lower commitment at the same spacing. The aligned histogram shows how annotated inter-PAS spacings fall across these distances (Figure 6B - lower).

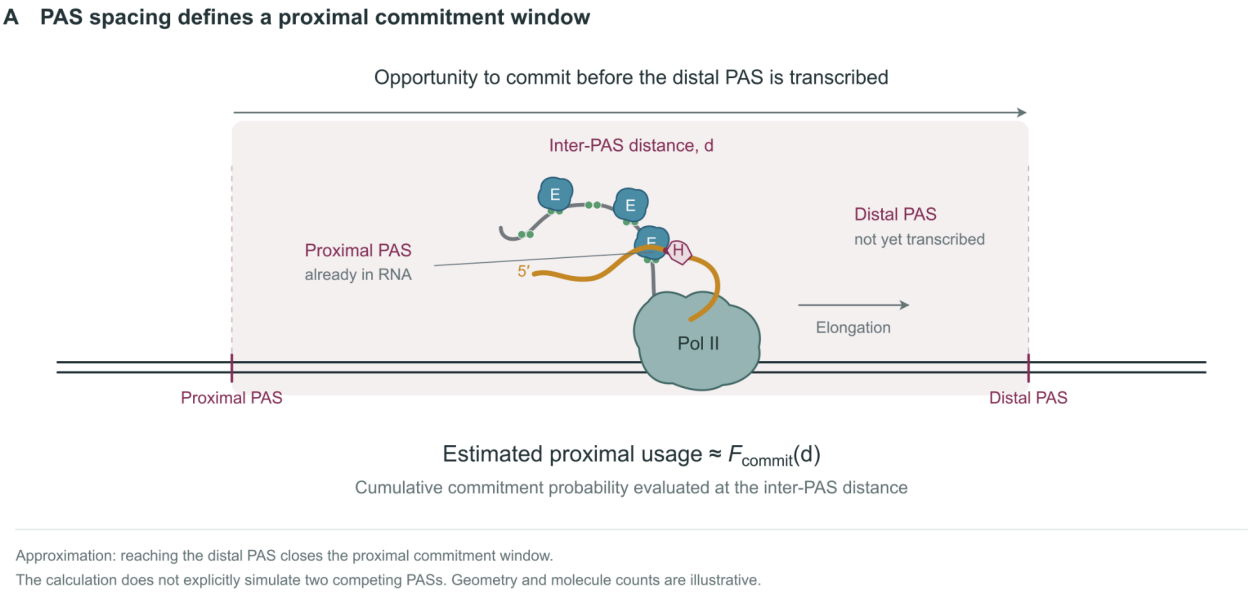


Figure 6A. The interval between PASs provides an opportunity for proximal commitment.

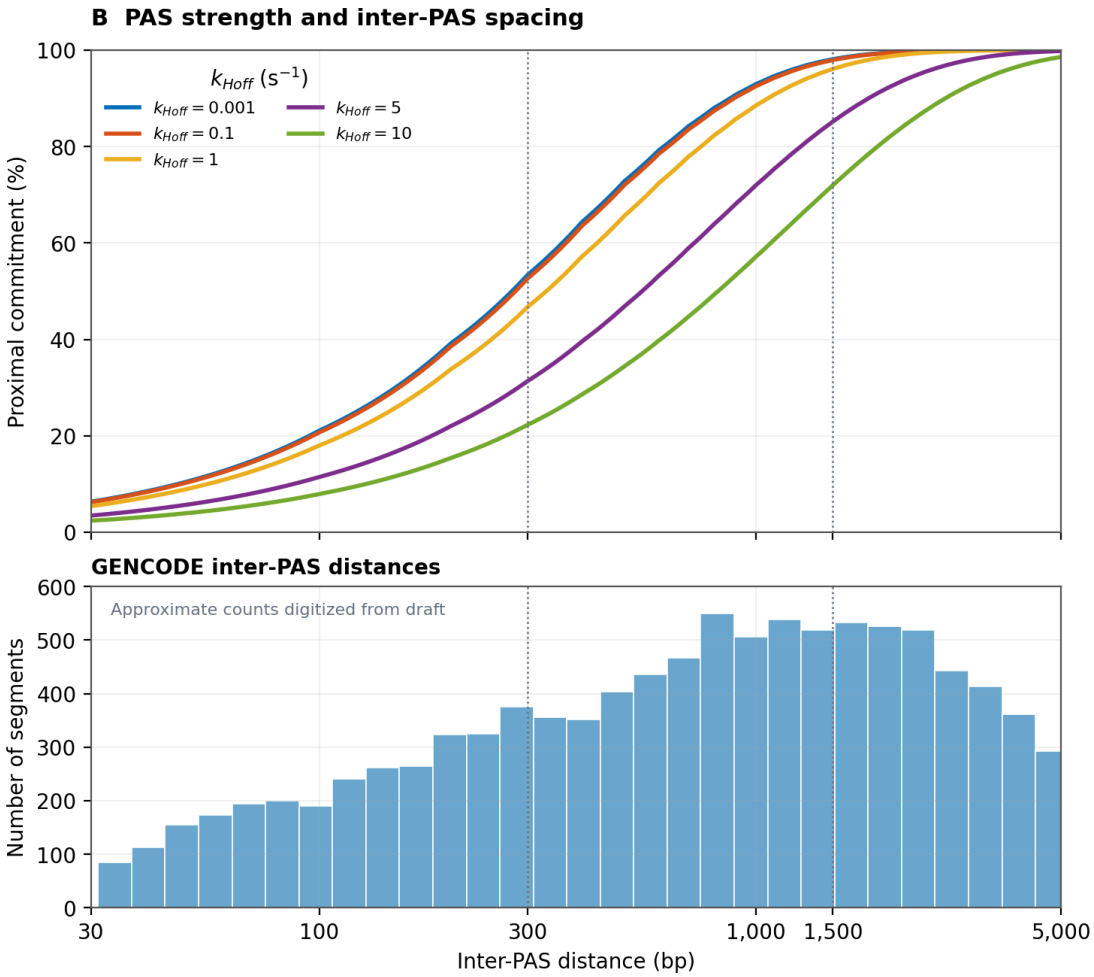


Figure 6B. Proximal commitment for kHoff = 0.001, 0.1, 1, 5, and 10 s^{-1}, with M = 5 and a 50 kb simulated gene. Both panels share a logarithmic axis spanning 30–5,000 bp.

Discussion

Discussions derived from the RESULTS sections:

- Chao's results are used roughly for comparison; they should not be the absolute benchmark.

- Why we chose M = 5: increasing M beyond 5 gains little efficiency

- The chosen parameter space is represented in a physiologically realistic manner. We utilized direct measurements and also derived all estimates based on published experimental data and literature.
    - For parameters that we could not find published data, we carefully implemented model approximations to quantitatively estimate the unknown parameters (i.e., kc) or utilized parameter values that qualitatively simulate experimental data from which we do not have a well-defined quantitative metric (i.e., phosphorylation dynamics across gene length).

- We need to state how the CPA factor/Pol II subunit abundance measurements translate into the numbers of E-factor versus the **number of actively transcribing Pol II** - we can convert the abundance from Joshua into the predicted CPA numbers using the billion protein molecules per cell estimate. We also need to be clear that the abundance of Pol II core does not necessarily mean that this correlates with actively transcribing Pol II because only a fraction of Pol II will be transcribing within the cell.
    - https://doi.org/10.1083/jcb.200206019 - Kimura et al. (2002) - ~20 (HeLa) to 25% (mammalian) of the total Pol II pool are actively transcribing or transiently immobilized (but engaged with DNA template - aka paused) in human genes
    - 10.1091/mbc.9.6.1523 - Jackson et al. (1998) - "each (HeLa) cell contains ~75,000 active polymerases II/III"
    - 10.3389/fgene.2012.00221 - Reider et al. (2012) ~65,000 active polymerase molecules
    - https://doi.org/10.1023/A:1024926710797 - Martin & Pombo (2003) - 65,000 to 85,000 in Hela cell. Also states that active pol II is ~25% of total Pol II pool within HeLa cells based on studies Jackson et al. (1998) and Kimura et al. (1999).

- Simply raising the abundance of E-factors (PAS-recognizing CPA factors) does not necessarily allow the CPA machinery assembly to be more efficient/lower CAD within the cell.
    - We show that even if the cell is able to raise E-factor abundance… , the CAD is decreased by … . raising this abundance leads to values that are likely to be physiologically unreasonable (that is, unattainable E-factor production by the cell.)

Addressing Open Questions Section:

- Why do certain CPA factors spatiotemporally associate with the Pol II CTD during certain portions of transcription?
    - Even though there is significant experimental evidence showing that E-factors have the tendency to associate with Pol II CTD early within transcription when compared to L-factors, there is no clear reason why L-factors cannot bind to the CTD before E-factors since both utilize Ser2p sites.
    - How does this impact “intermediate” CPA factors?
        - CPSF73 has the propensity to bind to the Pol II CTD intermediately within the transcription process and is vital for the cleavage function in the final assembly.

- How many Serine 2 phosphorylated heptamers are required for CPA factors to effectively bind the Pol II CTD?

- Is the Pol II CTD phosphorylated in a sequential or distributive manner? Is phosphorylation of the serine 2 positions processive or can these phosphorylation dynamics be described through unspecific position binding and unbinding events?

- Why do the majority/typical protein coding genes have a limited length distribution and not a larger range?
    - Serine 2 level could potentially/partially explain the both minimum and maximum gene length limit within cells/organisms. (If a protein coding gene is too short, not enough Ser2p sites on Pol II CTD for proper CPA and post translational operations. If a protein coding gene is too long, perhaps there is a loss in APA and post translational modification fidelity - that is, the proximal PAS will be selected as an uncontested majority by the CPA machinery.)

    - Our model framework can be utilized to help answer even bigger questions like why do many organisms have a distribution of the number of chromosomes? (Could potentially be used to explain chromosome number variability of polyploids and different organisms.)

- Why does InterPAS distance vary greatly among different genes/transcripts?
    - Median = 1,521 nucleotides (when excluding below 30 nt)
    - Range [30, 10^6] nt
    - Regulators like RNA-binding proteins, antisense transcripts, etc. that have the potential to interact with or occlude PAS sites (especially the proximal - in the canonical/our model’s predictions sense) that themselves can be regulated can act as another regulatory layer/mechanism that influences CPA assembly and CPA machinery “choice” for a PAS when multiple PAS are available via APA-associated transcripts.

- The cleavage/commitment rate (kc) is very simple for our model - kc represents only two reversible binding steps with a third irreversible step (for L). In reality there exist more complex binding events (i.e., more than those that involve just the Late CstF factors) and involve processing time for both the cleavage and polyadenylation processes to occur.

- Our Model makes a simplifying assumption that cleavage causes the transcribing Pol II to release from the DNA template. In reality this is not the case, but we use it here to quantify the assembly distance/time to quantify CPA assembly efficiency.
    - Our model can be expanded to include the downstream processes that influence PAS-dependent transcription termination.

Discussion Points on Model Limitations + Improving Model Results:

- Binding kinetics of CPA factors to Pol II CTD and between CPA factors remain undetermined.
    - These are critical values for creating more detailed/realistic future models that describe an accurate assembly process. Such values have the potential to significantly change the described CPA assembly efficiency provided by our model.

- Can Chip-chip be modified to quantify the number of heptads that are phosphorylated on transcribing Pol II in an accurate manner? This would be immensely helpful in quantifying Pol II CTD dynamics and for parameterizing this and future models.

    - **Does CHip-chip represent the % of serine 2 activated on the Pol II CTD at many heptad positions or a single heptad? How are multiple heptamers counted?**
        - *From our understanding Chip-chip takes Pol II at a certain position of DNA, says if it is phosphorylated at a given CTD amino acid via a measured signal - but does not provide the level nor number of phosphorylated CTD sites. This Ser2p signal "across the gene" is normalized via the "total Pol II at the gene positions" signal to account for Pol abundance changes in the gene due to pausing, elongation rate differences, etc. Thus, this Ser2p signal serves as a metric for a relative increase in Ser2p from the TSS to TTS.*